Hussein Saab

647-528-7842 | husseinsaab14@gmail.com | linkedin.com/in/huss-saab/ | github.com/husaab

Education

Toronto Metropolitan University (*formerly Ryerson University*) Toronto, Ontario
Bachelor of Science in Computer Science, Co-op *Sep 2022 – Dec 2027*

- Faculty of Science Deans List: 2022 – Present

Experience

Software Engineer Intern Sept 2026 – Present
Shopify *Toronto, ON*

- Developed billing and subscription features for Shopify merchants on the Merchant Subscriptions team, working across a Ruby on Rails monolith, GraphQL APIs, React/TypeScript frontend, MySQL, Kafka, and BigQuery
- Architected a cost-tracking system for an AI-powered WhatsApp merchant agent, attributing LLM token usage and messaging fees across **6,000+ conversations** and **200,000+ messages**, enabling granular cost visibility
- Lead the migration of Plus funnel analytics from a deprecated event pipeline to Event Refinery, dual-writing events across **84+** callsites to preserve **55K+** monthly events from **12K+** shops with zero downtime

Software Engineer Intern May 2026 – Aug 2026
Autodesk *Toronto, ON*

- Built TypeScript microservices and AWS CDK infrastructure for Autodesk's multi-region e-commerce payment systems using Lambda, DynamoDB Global Tables, SQS, EventBridge, Route 53 failover, and CloudWatch
- Developed and deployed a cloud-hosted LangGraph investigation agent using AWS CDK and CloudFormation, integrating Splunk, Dynatrace, CloudWatch, GitHub, and Atlassian MCP tools to correlate logs, traces, deployments, and source code
- Deployed the agent to customer-support Slack channels, helping on-call engineers resolve **30+ critical e-commerce incidents**, proactively detecting **44+ failed purchases** before escalation, and reducing investigation time by **60%**

Software Engineer, Full Stack Intern Jan 2026 – April 2026
Super.com *San Francisco, CA*

- Shipped end-to-end hotel booking and fintech features, including higher initial cashback, multi-room bookings, and Hopper Deals, contributing to **$5M+ in revenue**
- Owned features from TypeScript and XState frontends through Flask and gRPC backends, instrumenting and rolling out experiments with Amplitude and monitoring with Datadog
- Planned and executed secure consolidation of two active PostgreSQL booking tables with **7M+ records** using incremental migration strategies (dual writes, backfill) with zero downtime

Software Engineer, Backend & Automation Intern May 2024 – Aug 2025
IBM *Toronto, ON*

- Developed access-control microservices for IBM Supply Chain Intelligence Suite using Java, Spring Boot, Kafka, Redis, and Cassandra. Built access-control services for user groups, permissions, and secure feature access
- Deployed a Kafka monitoring service with Kubernetes and Docker, tracking **5M+ daily events** across **10+ microservices** and reducing manual monitoring by **40%**
- Built and optimized REST APIs with caching, pagination, retry logic, and JWT authentication, reducing response times by **65%**, while automating unit, integration, and API testing with JUnit, Mockito, Karate, and Jenkins

Projects

SchoolMule: AI Student Management | *TypeScript, React Native, Node.js, PostgreSQL, Python* **schoolmule.ca**

- Built a full-stack school platform with Next.js, Node.js and PostgreSQL, plus native **iOS and Android** apps in React Native, covering grades, report cards, attendance, parent–teacher messaging and classroom work, reducing administrative workload by **2,000+ hours annually**
- Orchestrated adoption by **4 underfunded private schools** serving **2,000+ students**; managed AWS deployment, migrated legacy student data, and delivered training and documentation to **150+ teachers**

Technical Skills

Languages: TypeScript, Java, Python, JavaScript, SQL, C/C++, Bash, HTML/CSS
Frameworks & Libraries: Spring Boot, React, Next.js, Node.js, Express, Flask, XState, SQLAlchemy
Cloud & Infrastructure: AWS, AWS CDK, Kubernetes, Docker, GitLab CI/CD, Jenkins
Data & Messaging: PostgreSQL, DynamoDB, Cassandra, Redis, Kafka, SQS, EventBridge, gRPC, Protocol Buffers
Observability & Tools: CloudWatch, Splunk, Datadog, Dynatrace, Amplitude, Git, Linux/UNIX